

E-Commerce Customer Segmentation & Cohort Retention

Executive briefing and retention playbook

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1. Executive Summary

The project brief describes a **25% rise in Customer Acquisition Cost (CAC)** over 12 months while marketing still relies on untargeted mass promotions. The question is where customers drop off after their first order and which customers are worth investing in.

The analysis covers **1,067,371 raw transaction lines** from Dec 2009 – Dec 2011. After cleaning, **805,549 lines** from **5,878 registered customers** in 41 countries (£17.74M revenue, 36,969 orders) were analysed with monthly cohort retention and RFM segmentation.

Period	Cleaned lines	Customers	Revenue
Dec 2009 – Dec 2011	805,549 (from 1.07M)	5,878 in 41 countries	£17,743,429.18
Month-1 retention	Not back in month 1	Champions + Loyal	At Risk revenue
23.1% (customer-weighted)	76.9% of new customers	84.7% of revenue from 43.1% of customers	£1.71M (9.6% of total)

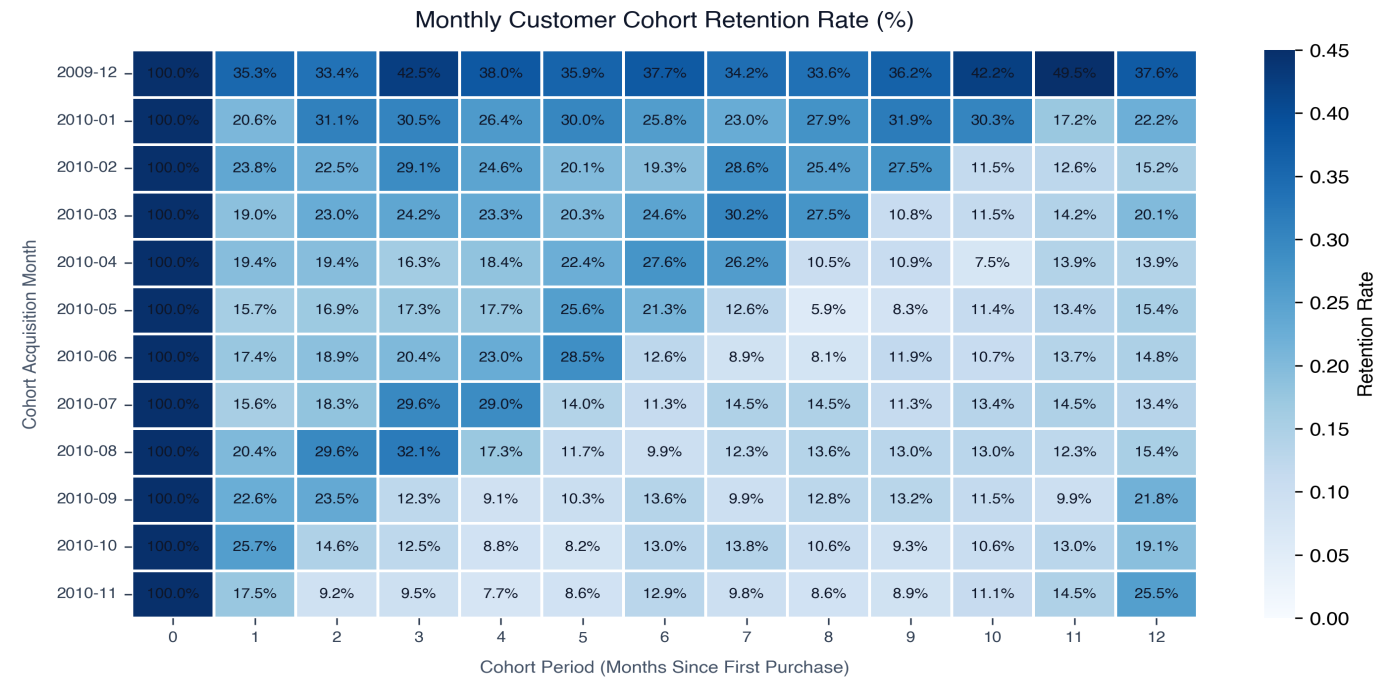
2. Data Preparation

The raw Online Retail II workbook (two yearly sheets) was processed in Python (src/etl_pipeline.py) and loaded into SQLite:

- **Cancellations and returns:** invoices starting with 'C' and lines with quantity ≤ 0 were removed.
- **Guest checkouts:** lines without a Customer ID were excluded, because cohort and RFM analysis need a persistent customer.
- **Invalid prices:** lines with a unit price $\leq$ £0 were removed.
- **Engineered fields:** line revenue (quantity $\times$ unit price), invoice month, cohort month (first purchase) and recency in days from 10 Dec 2011.

3. Monthly Cohort Retention

Customers are grouped by the month of their first purchase and tracked by how many buy again in each following month (month 0 to 24).



Findings

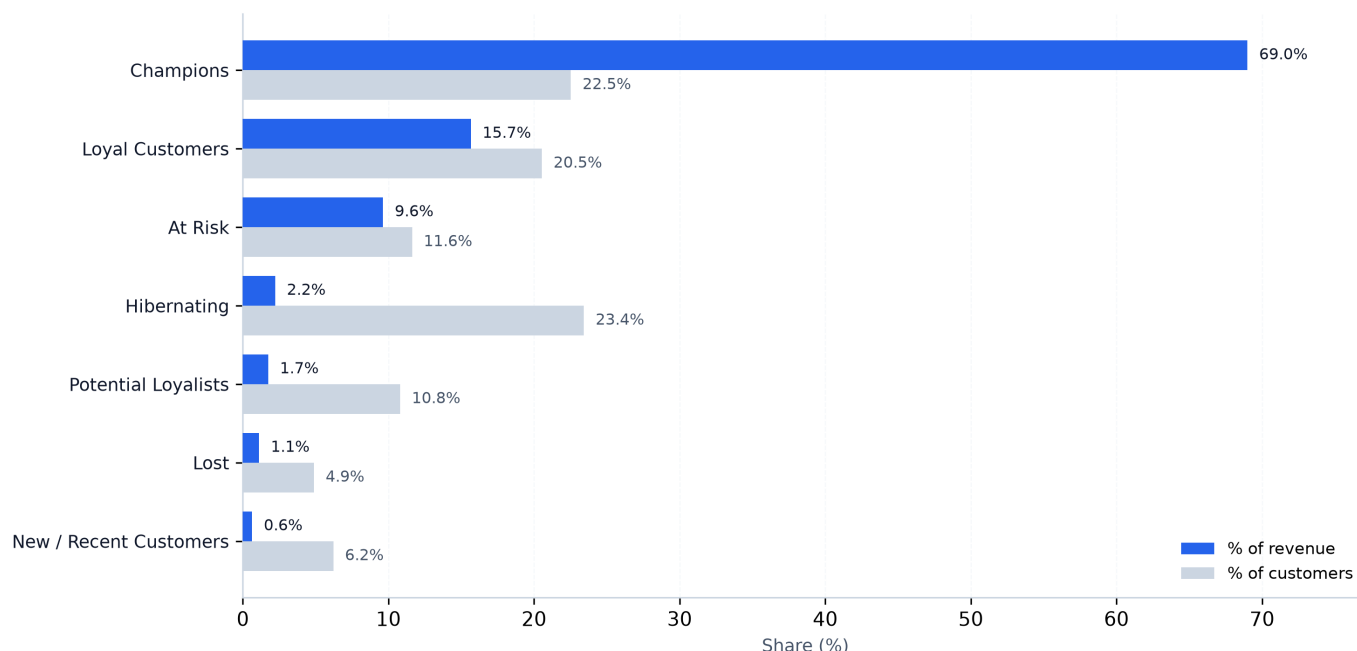
- **The month-1 cliff:** only **23.1%** of new customers buy again in the month after their first order, so 76.9% do not come back that month.
- **Flat after the cliff:** retention stays between **19.0% and 24.5%** from month 1 to month 12. The second purchase is the main lever.
- **Holiday cohorts retain worst:** month-1 retention for customers acquired in the Q4 peak was Nov 2010 17.5%, Dec 2010 9.2%, against the 23.1% average, consistent with promotions attracting one-off buyers.
- **Action window:** follow-up has to happen within the first 30 days, before the next month starts.

4. RFM Segmentation

Each customer is scored 1 to 5 (quintiles) on **Recency** (days since last order), **Frequency** (distinct orders) and **Monetary value** (total spend), then assigned to one of seven segments.

Segment	Customers	% base	Revenue	% rev	Avg recency	Avg orders	Avg spend
Champions	1,324	22.5%	£12,242,058	69.0%	20 days	16.9	£9,246
Loyal Customers	1,207	20.5%	£2,778,210	15.7%	74 days	5.8	£2,302
Potential Loyalists	635	10.8%	£309,190	1.7%	84 days	1.9	£487
New / Recent Customers	365	6.2%	£112,874	0.6%	29 days	1.4	£309
At Risk	683	11.6%	£1,708,453	9.6%	364 days	5.3	£2,501
Hibernating	1,377	23.4%	£394,013	2.2%	466 days	1.2	£286
Lost	287	4.9%	£198,631	1.1%	396 days	2.3	£692

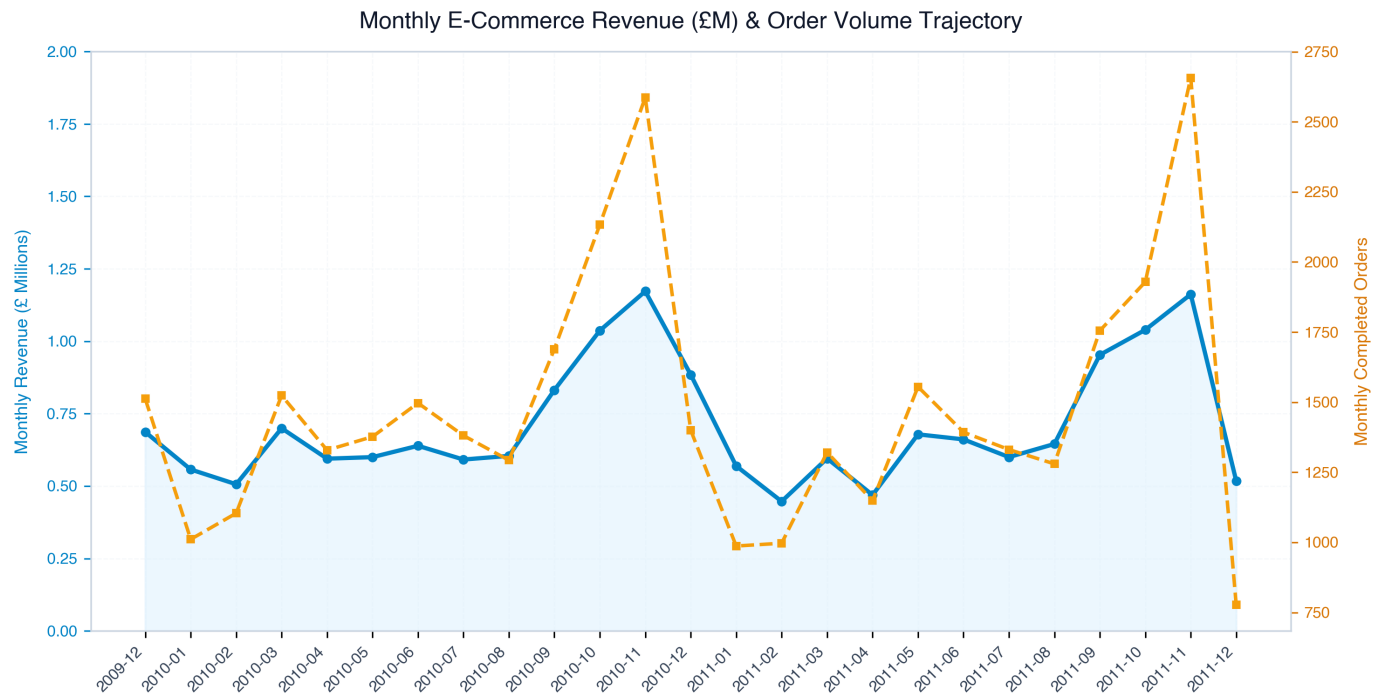
Revenue share vs customer share by RFM segment



Findings

- **Revenue is concentrated:** Champions are 22.5% of customers but 69.0% of revenue. With Loyal Customers, the top two segments (2,531 customers, 43.1%) bring in 84.7% (£15.02M).
- **High-value customers at risk:** 683 At Risk customers averaged 5.3 orders and £2,501 spend, but last ordered about 364 days ago on average. Together they represent **£1.71M** of historical revenue worth winning back.
- **Large dormant tail:** Hibernating and Lost customers are 28.3% of the base but only 3.3% of revenue. Paid remarketing to them is unlikely to pay back; low-cost email is enough.

5. Monthly Revenue and Seasonality



- **Strong Q4 seasonality:** revenue climbs from September and peaks in November in both years at around £1.2M a month, roughly twice the Q1 level.
- **December 2011 is partial:** the data ends on 9 December 2011, so the final drop is not a real decline.
- **Implication:** Q4 has the most active customers but the holiday cohorts retain worst, so a January re-engagement flow for November and December first-time buyers matters most.

6. Retention Playbook

Recommended actions by owner. Each should be run as a controlled test (holdout group) before full rollout.

A. Head of Marketing / CMO: budget and margin

1. **Shift budget from acquisition to retention:** move part of the broad paid-media budget into lifecycle email/SMS for New and At Risk customers.
2. **Stop blanket discounts:** keep discounts for targeted At Risk win-back offers only.
3. **Reward Champions without margin:** early access, exclusive products and service perks instead of price cuts.

B. CRM & Retention Lead: lifecycle triggers

1. **14-day post-purchase flow:** personalised recommendations based on the first order's category, to close the month-1 gap.
2. **60-day inactivity trigger:** start re-engagement when a customer passes 60 days without an order, well before the At Risk level (~364 days on average).
3. **Channel by value:** SMS/WhatsApp for high-value At Risk customers, a regular newsletter for Loyal and Potential Loyalists.
4. **Refresh RFM scores weekly** so customers moving between segments are caught early.

C. Product & Growth Manager: habit and tracking

1. **Loyalty tiers** with milestones at the 3rd and 5th order to move Potential Loyalists up.
2. **One-click reorder** from order history for frequently repurchased items.
3. **Track retention in the Power BI dashboard** (Overview, Cohort Retention, RFM Segments) to monitor month-1 retention and segment shifts.

7. Technical Architecture

- **SQLite database** (data/online_retail_analytics.db, built by the ETL script): fact_transactions, dim_customers_rfm, fact_cohort_activity and analytical views.
- **SQL** (sql/): schema and views, cohort retention query, RFM scoring with NTILE(5).
- **Power BI** (powerbi/): star-schema tables, DAX measures, theme and build guide for a three-page dashboard.
- **Tableau** (tableau/online_retail_analytics.tds): data source for a live SQLite connection via ODBC.
- **Report figures** are computed in src/report_metrics.py, so the README, this report and the slide deck stay in sync.

Notes and limitations

- The December 2009 cohort includes customers who bought before the data starts, which inflates its retention (35.3% in month 1).
- Customer-weighted retention divides all returning customers by all cohort customers that reached that month, so large cohorts weigh more.
- Recommendations are hypotheses to test; the data does not include marketing spend, so CAC impact cannot be measured here.